

Content-Based Filtering: Using Features

In this approach, we assume we *have* specific features for each item (movie) and we use these features to predict a user's preference.

The Setup

- n_u : Number of users.
- n_m : Number of movies (items).
- n : Number of features per movie.
 - Example: $n = 2$, where x_1 = Romance score, x_2 = Action score.

We treat this like a **Linear Regression** problem, but with a twist: we fit a separate linear regression model for **each individual user**.

The Model

For every user j , we learn a set of parameters $w^{(j)}$ and $b^{(j)}$.

To predict the rating user j would give to movie i :

$$\text{Prediction} = w^{(j)} \cdot x^{(i)} + b^{(j)}$$

- $x^{(i)}$: The feature vector for movie i (e.g., [0.99, 0] for a romance movie).
- $w^{(j)}$: The user's "preference" vector (e.g., [5, 0] if they love romance and hate action).
- $b^{(j)}$: The user's bias term.

The Cost Function (For One User j)

We want to minimize the squared error between the predicted rating and the actual rating, summing only over movies the user has actually rated ($r(i, j) = 1$).

$$J(w^{(j)}, b^{(j)}) = \frac{1}{2} \sum_{i:r(i,j)=1} ((w^{(j)} \cdot x^{(i)} + b^{(j)}) - y^{(i,j)})^2 + \frac{\lambda}{2} \sum_{k=1}^n (w_k^{(j)})^2$$

- The first term is the **Mean Squared Error** (prediction vs. actual).
 - The second term is the **Regularization** term (to prevent overfitting).
 - Note: The normalization constant $1/m^{(j)}$ is removed for mathematical convenience, as it doesn't change the optimal parameters.
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The Full Cost Function (For All Users)

To learn the preferences for *all* users simultaneously, we sum the cost function over all users ($j = 1$ to n_u).

$$J(\text{all } w, \text{all } b) = \frac{1}{2} \sum_{j=1}^{n_u} \sum_{i:r(i,j)=1} ((w^{(j)} \cdot x^{(i)} + b^{(j)}) - y^{(i,j)})^2 + \frac{\lambda}{2} \sum_{j=1}^{n_u} \sum_{k=1}^n (w_k^{(j)})^2$$

By minimizing this total cost function using **Gradient Descent**, we can learn the optimal parameters (w, b) for every single user, allowing us to predict their ratings for movies they haven't seen yet.

The Problem

This method works great **IF** we have the features $x^{(i)}$ (Romance score, Action score, etc.). But in the real world, we rarely have this detailed data for every single movie.